1. Clinical Question & Core Hypothesis

Clinical Question

PSMA-targeted therapies sometimes fail despite target presence — why does PSMA availability vary and under what tumor-microenvironment conditions does PSMA downregulation translate into clinical non-response? 2) Tier 1 paths used: (a) 'Androgens → downregulates → PSMA' provides the core mechanistic axis (AR-mediated repression of target antigen); (b) 'PSMA → targets → Therapy' connects antigen level to therapeutic efficacy. I built on these because they give a direct mechanistic link from hormonal signaling to target availability and therapy action. 3) Tier 2 supporting evidence: 'Neuroendocrine hormones → treats → Prostate cancer' (indicates that endocrine/neuroendocrine modulation can shift androgen signaling and thus PSMA), and the deprivation-score edges (population/access factors) suggest real-world modifiers of treatment timing and selection that could confound observed responses. 4) Why therapy_response endpoint: the mechanistic chain links AR activity to PSMA expression which directly determines binding/efficacy of PSMA-directed agents, so a therapy-response proxy is the most relevant and actionable endpoint. 5) Deliberate exclusions: I excluded socioeconomic/deprivation complexity from the core claim to keep the mechanistic focus on tumor biology; deprivation_score and broader 'therapy access' effects are listed as secondary_modifiers for follow-up analyses rather than part of the mechanistic hypothesis itself?

Working Hypothesis

In breast cancer, androgen receptor (AR) signaling modulates PSMA (FOLH1) expression, such that AR activation downregulates PSMA density on tumor cells and/or tumor-associated vasculature. This AR-driven downregulation reduces the binding and efficacy of PSMA-targeted therapies. The magnitude of this effect is modified by the tumor immune architecture, specifically TLS-like lymphoid aggregates: TLS-low tumors exhibit stronger functional consequences of AR-mediated PSMA suppression, leading to the poorest therapy response to PSMA-directed agents; TLS-high tumors may partially buffer this effect through enhanced local immune organization and potentially higher baseline target accessibility. Consequently, tumors with AR-high and TLS-low status will show the lowest therapy response to PSMA-directed treatment, whereas AR blockade or therapies that upregulate PSMA, particularly in TLS-rich contexts, will restore target density and improve therapeutic efficacy. The imaging feature to anchor these predictions is TLS-like lymphoid aggregates, and the endpoints are surrogate/clinical measures of therapy response to PSMA-directed therapy (e.g., radioligand uptake or tumor response).

Element	Value
Imaging feature	TLS-like lymphoid aggregates
Anchor gene(s)	PSMA, AR
Primary endpoint	therapy_response

2. Why This Gene — Not Others

Primary hypothesis gene(s)

PSMA: PSMA-targeted therapies sometimes fail despite target presence — why does PSMA availability vary and under what tumor-microenvironment conditions does PSMA downregulation translate into clinical non-response? 2) Tier 1 paths used: (a) 'Androgens → downregulates → PSMA' provides the core mechanistic axis (AR-mediated repression of target antigen); (b) 'PSMA → targets → Therapy' connects antigen level to therapeutic efficacy.

AR: argeted therapies sometimes fail despite target presence — why does PSMA availability vary and under what tumor-microenvironment conditions does PSMA downregulation translate into clinical non-response? 2) Tier 1 paths used: (a) 'Androgens → downregulates → PSMA' provides the core mechanistic axis (AR-mediated repression of target antigen); (b) 'PSMA → targets → Therapy' connects antigen level to therapeutic efficacy.

Alternative / secondary genes considered

Tier	Gene	Role / Reason excluded from core
Secondary	Neuroendocrine_hormones	Supporting modifier from knowledge graph
Secondary	Deprivation_score	Supporting modifier from knowledge graph

3. Clinically Testable Predictions

Prediction 1: expression has been observed in subsets of breast cancer and especially within tumor vasculature in some contexts, making PSMA a potential target beyond prostate cancer. AR signaling is active in a subset of breast cancers (notably ER-positive subtypes) and can regulate gene expression programs.

Prediction 2: In functional experiments, restoring or enhancing the anchor gene's signaling will improve immune-mediated tumor control in imaging-feature-positive models.

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Novel mechanistic hypothesis derived from knowledge graph analysis.

Caveats & Risks

- Hypothesis is computationally derived and requires experimental validation.

Alternative Directions

- Neuroendocrine_hormones: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.
- Deprivation_score: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.